

SPOTIFY SONG POPULARITY PREDICTION USING MACHINE LEARNING

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Dataset

- ❖ Name: Spotify Tracks Dataset
- ❖ Source: Kaggle
- ❖ Rows: ~50,000+ (depends on dataset version)
- ❖ Columns: ~15+ audio features

Source Link

<https://www.kaggle.com/datasets>◆

<https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>

Target Variable

- Popularity (song popularity score)

Task Type

- Regression (Supervised Learning)

Problem Statement

To predict Spotify song popularity using audio features such as danceability, energy, tempo, and valence, and evaluate the performance of different machine learning models in accurately predicting song popularity.

Research Questions

1. Baseline Model Performance

How effectively do baseline machine learning models predict Spotify song popularity using audio features?

2. Model Comparison

How does the performance of different regression models compare in predicting Spotify song popularity?

3. Effect of Preprocessing

What is the impact of preprocessing techniques such as missing value handling and scaling on prediction accuracy?

4. Feature Importance

Which audio features have the greatest influence on Spotify song popularity, and how can their impact be interpreted?

5. Evaluation Metrics Impact

How does the choice of evaluation metrics (MAE, MSE) influence model comparison results?

6. Model Robustness

How stable and reliable are the models under different data splits and testing conditions?

7. Final Model Selection

Which machine learning model is most suitable for Spotify song popularity prediction considering accuracy, interpretability, and performance?

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Methodology

- ❖ Data loading and exploration
- ❖ Data cleaning (handling missing values and duplicates)
- ❖ Feature selection
- ❖ Train-test split
- ❖ Model training and evaluation
- ❖ Visualization of results
- ❖ Model comparison

Evaluation Metrics

- ❖ MAE (Mean Absolute Error)
- ❖ MSE (Mean Squared Error)
- ❖ R^2 Score (Coefficient of Determination)

Expected Outcome

The project is expected to identify key audio features influencing song popularity and determine the most effective machine learning model for prediction.